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Towards precision agriculture tea leaf disease detection using CNNs and image processing

Irfan Sadiq Rahat[1], Hritwik Ghosh[1], Suresh Dara[1]✉ & Shashi Kant[2]✉

In this study, we introduce a groundbreaking deep learning (DL) model designed for the precise task of classifying common diseases in tea leaves, leveraging advanced image analysis techniques. Our model is distinguished by its complex multi-layer architecture, crafted to adeptly handle 256 × 256 pixel images across three color channels (RGB). Beginning with an input layer complemented by a Zero Padding 2D layer to preserve spatial dimensions, our model ensures the retention of crucial geographical information across its depth. The innovative use of a convolutional layer with 64 7 × 7 filters, followed by batch normalization and Rel U activation, allows for the extraction and representation of intricate patterns from the input data. Key to our model's design is the incorporation of residual blocks, facilitating the learning of deeper networks by alleviating the vanishing gradient problem. These blocks combine Conv2D layers, batch normalization, activation layers, and shortcut connections, ensuring robust and efficient feature extraction at various levels of abstraction. The GlobalAveragePooling2D layer towards the model's end succinctly summarizes the extracted features, preparing the model for the final classification stage. This stage features a dropout layer for regularization, a dense layer with 512 units for further pattern learning, and a final dense layer with 8 units and a soft max activation function, producing a probability distribution across different disease classes. Our model's architecture is not just a testament to the sophistication of modern deep learning techniques but also highlights the novelty of applying such complex structures to the challenges of agricultural disease detection. We utilized a datasets consisting of 4000 high-resolution images of tea leaves, encompassing both diseased and healthy states, meticulously captured in the tea gardens of Pathantula, Sylhet, Bangladesh. Employing the Canon EOS 250d Camera ensured detailed representation crucial for training a robust deep learning model for disease detection in tea plants. By achieving remarkable accuracy in identifying diseases in tea leaves, this research not only sets a new benchmark for precision in agricultural diagnostics but also opens avenues for future innovations in the field of precision agriculture.

Keywords Tea leaf disease classification, Image analysis for plant health, Agricultural disease detection with AI, Precision agriculture through machine learning, Machine learning optimizers in agriculture

Tea derived from the leaves of the Camellia sinensis plant, holds a prestigious position as a widely consumed beverage and a significant agricultural product, particularly in Bangladesh. The inception of tea cultivation in this region dates back to the nineteenth century, with the establishment of the first tea garden in Chittagong in 1840. The cultivation of tea in Bangladesh spans across diverse ecological zones, including the Surma valley in the greater Sylhet region, the Halda valley in Chittagong, and the Karatoa valley in Panchagarh. Presently, the country is home to 162 tea estates, covering an expanse of approximately 60,179 hectares and producing an annual output of nearly 67.38 million kg of finished tea[1]. This results in an average production rate of about 1270 kg per hectare, contributing a significant 0.11% to the national GDP. The Bangladesh Tea Research Institute (BTRI) has been instrumental in this sector, releasing numerous high-yielding and quality tea clones, as well as hybrid seed stocks[1]. Tea holds a special place in the daily life of Bangladeshis, being a staple in their daily routines and contributing significantly to the country's economy as the second-largest export-based cash crop, accounting for 1% of Bangladesh's GDP.

Despite its prominence, the tea industry in Bangladesh faces substantial challenges due to various pests and diseases, influenced by the unique environmental conditions of the region. These challenges necessitate the

[1]School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India. [2]Department of Management, College of Business and Economics, Bule Hora University, Blue Hora, EthiopiaAfrica. ✉email: darasureh@live.in; skant317@gmail.com

development of innovative and efficient disease detection methodologies[2,3]. In response to these challenges, this research delves into the capabilities of machine learning and deep learning, aiming to transform the landscape of tea leaf disease detection. Utilizing a rich datasets of tea leaf images from the verdant tea gardens of Sylhet, the study addresses the prevalent diseases that impede tea production. The deep learning model proposed in this research, with its intricate design and the application of various optimizer, showcases promising results in tackling these agricultural challenges. Notably, the Adam optimizer has proven to be exceptionally effective, delivering an impressive accuracy of 99%. This work delves into the critical role of agriculture in bolstering the national income of countries like India and addresses the significant impact of plant diseases on production quality and quantity, leading to economic losses. It highlights the utilization of computer vision and soft computing by researchers[1] to automate plant disease detection using leaf images, summarizing various studies along with their advantages and drawbacks. The study provides an in-depth comparison of hand-crafted and deep learning-based feature extraction methods, exploring their strengths and weaknesses in identifying disease symptoms through image features like color, texture, and shape across diverse crops. This research not only underscores the potential of deep learning in agricultural applications but also lays the groundwork for future innovations in precision agriculture, steering towards more sustainable and efficient farming practices. The proposed model, as illustrated in Fig. 1 begins with an Image Dataset that undergoes Data Annotation to label the relevant features. This annotated data is then enhanced through Data Augmentation, a technique to artificially increase the size and diversity of the dataset by creating modified versions of the images. The annotated and augmented data then enters the core of the model, which is divided into three main stages: Training, Validation, and Testing. During the Training phase, the model uses the data to learn and adapt its parameters, specifically for Disease Training. The Validation process ensures that the model generalizes well to new data, preventing over-fitting. The Testing phase assesses the model's performance in predicting or detecting diseases. The results from the testing phase are then subjected to Performance Verification, ensuring that the model's outputs are accurate and reliable. The final step is Disease Detection, where the model, likely built on a Convolutional Neural Network (CNN) framework, identifies diseases based on the patterns it has learned.

As part of our comprehensive research endeavor to revolutionize tea leaf disease detection using advanced deep learning techniques, we employed a questionnaire-based approach for data collection. The questionnaire served as a crucial tool to gather valuable insights and perspectives from tea farmers, agronomists, and other stakeholders directly involved in tea cultivation and[4] disease management practices. The primary purpose of the questionnaire was to solicit firsthand information regarding the prevalence, severity, and management of tea leaf diseases in the region, particularly in the tea gardens of Sylhet, Bangladesh. By[5] engaging with local experts and practitioners through structured surveys and interviews, we aimed to gain a deeper understanding of the challenges faced in disease identification and management within the tea industry. The relevance of the questionnaire to our study lies in its ability to provide qualitative data that (Fig. 1) complements the quantitative analysis derived from image datasets and model performance metrics. The insights gleaned from the questionnaire responses enriched our understanding of the real-world complexities associated with tea leaf diseases, thereby informing the development and validation of our deep learning model.

The primary contribution of this research lies in the development of a deep learning-based model optimized for high-precision tea leaf disease detection. Unlike previous works that rely on traditional machine learning techniques or pre-trained models, our approach employs a custom convolutional neural network (CNN) architecture with enhanced feature extraction capabilities. We introduce a Zero Padding 2D layer and residual blocks to improve classification accuracy while reducing computational overhead. Furthermore, our dataset comprises 4,000 high-resolution images captured from tea gardens in Sylhet, Bangladesh, ensuring domain-specific robustness. The results demonstrate superior performance compared to existing methods, highlighting the potential of our model for real-world agricultural applications.

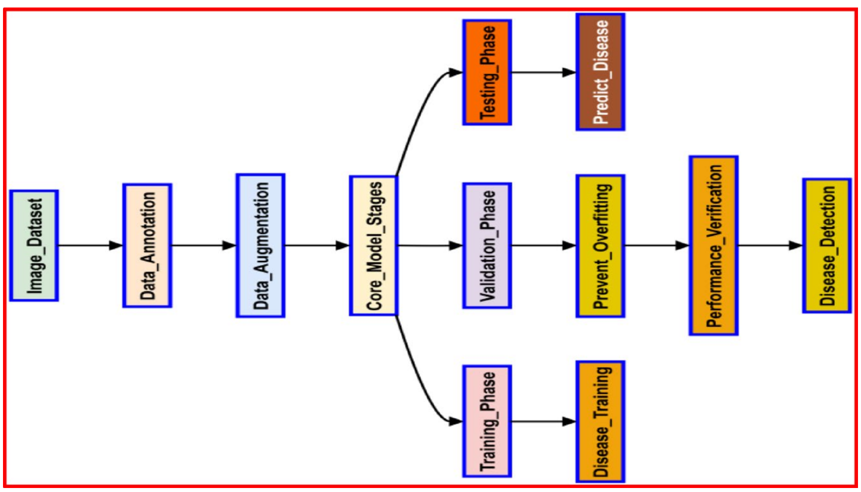

Fig. 1. Advanced deep learning-based tea leaf disease detection process.

The remainder of this paper is structured as follows: Section "Related work" provides a review of related works in tea leaf disease detection and deep learning approaches in precision agriculture. Section "Data acquisition" details the proposed methodology, including dataset acquisition, preprocessing, and the CNN architecture used for classification. Section "Model performance" presents the experimental setup, evaluation metrics, and results analysis. Section "Results and discussion" discusses the findings, limitations, and potential improvements. Finally, Section "Conclusion" concludes the paper and outlines future research directions.

Contributions of this study

- Innovative Model Architecture: This research introduces a deep learning model that is specifically tailored for the classification of tea leaf diseases. Unlike traditional methods, our model integrates a multi-layer convolutional neural network (CNN) with advanced image processing techniques. The architecture, designed to handle high-resolution images, ensures precise disease detection while retaining the important spatial features inherent in tea leaf images.
- Application of Residual Blocks: A novel contribution of this study is the use of residual blocks within the CNN architecture. These blocks help alleviate the vanishing gradient problem and enable the model to train deeper networks effectively, leading to better feature extraction and enhanced classification accuracy. This approach has not been widely applied in the context of tea leaf disease detection, highlighting the originality of the model.
- Optimization for Precision Agriculture: Our study also employs advanced optimization techniques such as the Adam optimizer, which significantly improves the training efficiency and final accuracy of the model. By achieving an impressive 99% accuracy in classifying tea leaf diseases, the model stands as a benchmark for precision agriculture in tea plantations, providing farmers with a tool for early disease detection and intervention.
- Comprehensive Dataset from Sylhet, Bangladesh: This research uses a unique dataset consisting of 4000 high-resolution images of tea leaves from the Pathantula Tea Garden in Sylhet, Bangladesh. These images encompass both diseased and healthy states, ensuring that the model can learn from a diverse range of real-world scenarios. The dataset was specifically captured with the Canon EOS 250d Camera, allowing for detailed representation crucial for robust deep learning model training.
- Impact on Tea Industry and Future Research: The findings of this research not only contribute to the scientific community but also provide practical solutions for the tea industry. The model can be integrated into real-world applications, such as automated disease detection systems, helping farmers to manage crop health more efficiently. Moreover, the research lays the foundation for future work in precision agriculture by incorporating multi-spectral imaging and real-time disease detection models for other agricultural sectors.

Background study

Tea, derived from the leaves of the Camellia sinensis plant, is a globally significant agricultural product, especially in Bangladesh, where it holds both economic and cultural importance. As one of the country's major export-oriented cash crops, tea contributes approximately 1% to Bangladesh's GDP and supports the livelihoods of thousands of farmers. However, the productivity and quality of tea are often threatened by various diseases such as Brown Blight, Red Leaf Spot, and Algal Leaf Spot, which can significantly impact yield and profitability. Addressing these challenges necessitates innovative solutions for early and accurate disease detection.

Traditional methods of detecting tea leaf diseases typically involve manual inspection by agronomists or farmers. While these approaches rely on expert knowledge, they are inherently time-consuming, labor-intensive, and prone to errors, especially in large-scale plantations. Moreover, the subtleties of early-stage disease symptoms are often difficult to detect with the naked eye, leading to delays in implementing remedial actions. Such limitations underscore the need for automated and precise disease detection systems.

Recent advancements in technology have opened new avenues for addressing agricultural challenges. Machine learning (ML) and deep learning (DL) have emerged as transformative tools in precision agriculture, enabling automated disease detection and classification with high accuracy. For instance, convolutional neural networks (CNNs) have been successfully deployed to identify diseases in crops like tomatoes, rice, and wheat, demonstrating their capability to process complex image data and detect subtle patterns. Techniques like transfer learning and data augmentation have further enhanced the performance of these models in scenarios with limited datasets.

Despite these advancements, limited research exists on leveraging deep learning specifically for tea leaf disease detection. Studies focusing on other crops have showcased the potential of DL models, but tea plantations present unique challenges, including variations in leaf texture, color, and disease presentation. The absence of robust, high-accuracy models tailored to the nuances of tea leaves highlights a critical gap in the literature.

This research aims to bridge this gap by introducing an advanced DL model specifically designed for classifying tea leaf diseases. By utilizing a dataset of 4,000 high-resolution images captured from the tea gardens of Sylhet, Bangladesh, the study seeks to establish a benchmark for precision in tea leaf disease diagnostics. Through this work, we aspire to contribute to the broader field of precision agriculture, empowering farmers with reliable tools for timely disease management and fostering sustainable tea cultivation practices.

Abbreviations and their full forms

To enhance clarity and comprehension, this section lists the abbreviations used throughout the paper along with their corresponding full forms. The table below serves as a quick reference for readers, ensuring consistency in understanding technical terms (Table 1).

Abbreviation	Full form
DL	Deep learning
CNN	Convolutional neural network
RGB	Red, green, blue
BTRI	Bangladesh Tea Research Institute
SVM	Support vector machine
SGD	Stochastic gradient descent
GAN	Generative adversarial network
CAD	Coronary artery disease
CT	Computed tomography
R-CNN	Region-based convolutional neural network
FCNN	Fully convolutional neural network

Table 1. List of abbreviations and their full forms.

Related work

Mamun[1] explores the complex world of Bangladeshi tea cultivation in his 2019 study. Tea is an important cash crop and export good. He highlights the Bangladesh Tea Research Institute's contribution to improving agricultural practices as he traces the industry's development from its founding in Chittagong in 1840 through a variety of ecological zones. Mamun emphasizes that, given the limited amount of land available, creative land use and farming techniques are essential to meeting domestic and export demands. The research, which accounts for 0.11% of Bangladesh's GDP, is an invaluable resource for comprehending the complexity and potential of the tea industry in the country.

In their 2018 study, Hossain et al.[2] discussed the serious effects of leaf diseases on Bangladesh's tea output and offered a cutting-edge image processing technique for early disease diagnosis. They created a method that uses a SVM classifier to accurately classify diseases such as algal leaf diseases and brown blight by utilizing eleven different features. In comparison to earlier research, their method reduced the feature set and produced a high classification accuracy of over 90% along with a 300 ms processing time reduction per leaf image. This development is expected to improve disease control effectiveness and raise Bangladesh's tea industry's standing internationally. Jahan et al.[3] examined the effects of extended usage of nitrogenous fertilizers on soil characteristics and tea leaf quality in five tea gardens in Bangladesh in their 2022 study. The study found that the application of nitrogenous fertilizers and soil pH had a negative relationship that reduced soil toxicity and nutrient deficits, which in turn affected tea quality and productivity. The study found that there has been a notable drop in soil pH and key nutrient availability, along with a loss in tea leaf quality, even if the pace of tea leaf output has been constant for the past 20 years. This study emphasises how important it is to use sustainable fertilization techniques in order to preserve soil health and guarantee Bangladesh produces high-quality tea. Using transfer learning and fine-tuning techniques, Ramdan et al.[6] address the difficulties of training DL models for tea illness diagnosis with less data in their 2020 work. They used smaller, task-specific datasets to test per-trained DCNN models, such as VGGNet, ResNet, and Xception, which were first trained on Image Net. To reduce task disparities, the models were adjusted by using different re-training and fine-tuning schemas for performance assessment. The results demonstrated the effectiveness of fine-tuning in small data-set contexts, as it outperformed transfer learning alone, partial fine-tuning, and training from scratch. However, the incorporation of fine-tuning alone shown limited effectiveness.

The LeNet Convolutional Neural Network (CNN) was utilized by Gayathri et al.[7] to identify tea leaf disease with the goal of improving diagnostic precision in India's thriving tea sector. Due to different phytopathological signs, the deep learning model performed especially well in diagnosing Blight leaf disease, reaching the greatest accuracy. The accuracy of the classification of other illnesses ranged from 84 to 93%, with Tea Red Scratching ranking 94%. This CNN model performed better than standard back-propagation networks, which produced an accuracy of 85% for the Blister Blight disease. This indicates the potential of DL in the management of agricultural diseases. In order to solve the serious problem of tea leaf illnesses in India, a major producer and consumer of tea, Srivastav et al.'s study[8] uses convolutional neural networks (CNNs) for disease diagnosis. India's economy greatly benefits from the tea sector, which also creates a large number of jobs, especially in rural areas. On the other hand, diseases that affect tea leaves can be detrimental to the crop's pace of growth and total production. The researchers created a CNN model with a set of leaf images, and then used the "ADAM" optimizer to show how effective the model was at detecting diseases with an accuracy of 84% and an error rate of 16%. Additionally, they proposed future directions for image augmentation approaches to improve the results of diagnostic procedures. An integrated learning-based technique was developed by Wang et al.[9] to detect pests and diseases on tea leaves, with a focus on Apolygus lucorum and Leaf blight. The study used YOLOv5 models that had been updated and improved with GAM and CBAM attention mechanisms. The Weighted Box Fusion algorithm was then used to merge the models' predictions. With an average accuracy of 79.3%, this method outperformed the results of separate models by 8.7% and 9.6%, respectively. Accuracy and model complexity are balanced in the suggested method, which promises enhanced illness identification in tea cultivation.

To solve a major difficulty in agriculture, Bhateja et al.[10] introduced a DL-based method for identifying and categorizing plant leaf diseases. Their model, which was trained on a publically accessible dataset, demonstrated 97.35% training accuracy and the ability to correctly identify up to 15 distinct plant diseases. This high degree of accuracy shows how the suggested system can be used in actual agricultural settings to help with early disease identification and crop disease prevention. In order to address the serious problem of liver illness, which is one of the main causes of death globally, Rathore et al.[11] used DL approaches for hepatic image processing. They suggested segmenting liver lesions in CT scans using a Fully Convolutional Neural Network (FCNN), which would make it easier to plan treatments, monitor clinical responses, and gauge the severity of the disease. The study's main objective was to differentiate between carcinogenic and non-cancerous tumors, a difficult task requiring a lot of resources and experience. Using CNN and AlexNet models, a binary classifier was created with the goal of distinguishing between livers in good condition and livers that had hemochromatosis. AlexNet performed more accurately than CNN, obtaining 95% versus 90%, demonstrating its efficacy in the identification of liver illness. India's agriculture industry, which is essential for feeding its 1.4 billion people, has a critical need for the exploration of variousML and DL techniques for potato disease detection.

Choudrie et al.[12] addressed this requirement with a thorough study. This study tested multiple approaches to find the best way to detect plant diseases using the Plant Village Dataset from Kaggle, which comprises many kinds of plant leaves, including potato leaves. With an astounding accuracy rate of 95–77%, the results demonstrated the exceptional performance of Convolutional Neural Networks (CNNs) and demonstrated its promise as a dependable tool for prompt and accurate plant disease identification.

Focusing on the crucial problem of disease-related losses in tomato crops, Kumar et al.[13] draw attention to the drawbacks of conventional disease detection techniques, which are costly, time-consuming, and necessitate specialized knowledge in plant pathology. The study offers a thorough analysis of deep learning methods for tomato crop disease detection and classification, looking at several architectures, their benefits, drawbacks, and performance. In order to help farmers with early-stage disease prediction and crop loss estimation, the manuscript emphasizes the need for better early disease detection methods, the development of preprocessing techniques specific to particular datasets, and the creation of user-friendly tools. These areas are highlighted as promising directions for future research. Rizvanov et al.[14] highlights the vital role of agriculture in India's GDP and the necessity of minimizing threats to crop production. The paper emphasizes the challenges of manually identifying crop leaf diseases, which can lead to misdiagnosis and reduced yields. To tackle this issue, the authors propose an automated system utilizing Convolutional Neural Networks (CNNs) for the classification and identification of vegetable leaf diseases. They utilize the Plant Village database, curating a new dataset specific to their study, comprising five classes of vegetable diseases with images of both diseased and healthy leaves. The use of DL more especially CNNs, for the detection of wheat rust disease—a common danger to global food security—is examined by Smys et al.[15]. While acknowledging the abundance of food produced globally, the study also draws attention to the serious concerns that pollinators, plant diseases, and climate change pose. Plant diseases can have disastrous consequences in Ethiopia, where small-scale farming is the main source of income for many communities. The authors emphasize the difficulties specialists have in enhancing the effectiveness of artificial intelligence detection methods given the complexity of crop diseases and point out that crop decrease is a significant global issue. The study presents a deep learning method that improves the model's disease detection performance by utilising the RGB values of the illness's colour on crops. The significant influence of plant diseases on agricultural productivity and a nation's economy is discussed by Jasim and AL-Tuwaijari[16]. Using photos from the Plant Village dataset, they provide a DL-based approach for categorising and identifying plant leaf diseases. They used a dataset of 20,636 plant pictures, covering both healthy and diseased leaves, with a focus on popular plants found in Iraq, such as potatoes, tomatoes, and peppers. They classified 15 classes using a Convolutional Neural Network (CNN), 12 of which were for various plant diseases and 3 of which were for healthy leaves. With testing results of 98.029% and training results of 98.29%, the system's remarkable accuracy confirmed its efficacy in disease identification and categorization.

The integration of artificial intelligence in agriculture is examined by Das et al.[17] who highlight the critical requirement for accurate disease identification in crops, particularly rice crops. They support deep learning methods, citing Mask Region-Based Convolutional Neural Network (Mask R-CNN) as the best option for precise disease area diagnosis and image segmentation. The study explores the application of image processing for the extraction of disease features, going over a number of methods such as contouring, edge detection, area computation, image enhancement, dimension analysis, correlation, and colour space assessment. The outcomes of the programming implementations highlight the substantial progress and future opportunities that DL and image processing offer the agriculture industry, with the goal of achieving nearly flawless precision in disease identification and treatment delivery. S, N. H. & H, S[18]. use ML methods, such as CNN, R-CNN, and fuzzy-SVM, to study tomato leaf disease detection. Using a variety of image processing techniques, the study classifies diseases; the R-CNN-based Classifier performs better than the others, with an accuracy rate of 96.735%. This study demonstrates how DL approaches might help farmers diagnose diseases earlier and minimise agricultural losses. To address the global effect of CAD-related mortality, Wahab Sait and Dutta created a deep-learning-based method in 2023[19] to diagnose Coronary Artery Disease (CAD) using CT scans. Their goal was to lessen the difficulties that come with using the large image datasets and high computing costs associated with the current CAD detection techniques. To improve the quality of the CT images, the researchers applied an image enhancement technique. They also used the You Look Only Once (YOLO) V7 for feature extraction and Aquila optimisation to adjust the UNet + + model's hyperparameters. The suggested approach outperformed recent methods with remarkable results in accuracy, recall, precision, F1-score, and other measures across two datasets. The findings imply that, despite its limitations, this model can be a useful tool for doctors diagnosing CAD.

With an emphasis on tomato plants, Chowdhury et al.[20] present a DL-based method for the automatic and trustworthy diagnosis of leaf diseases. They identify several tomato diseases by using 18,161 segmented and plain photos of tomato leaves coupled with a recently developed convolutional neural network called EfficientNet. The authors provide an analysis of the effectiveness of two segmentation models, namely U-net and Modified U-net, and showcase outcomes for binary, six-class, and ten-class categorization of leaf conditions. According to the study, segmented images and deeper networks improve classification performance. The Modified U-net and EfficientNet-B7 models achieve especially high accuracy for various classification categories. Their work demonstrates the promise of DL in agricultural disease management by outperforming previous studies in terms of performance. In their work, Rukhsar and Upadhyay[21] tackle the crucial problem of early rice leaf disease detection by identifying common rice diseases as Blast, Blight, and Tungro using transfer learning using DenseNet201. Compared to a more basic CNN model that only achieved 62.20%, their approach shows a notable boost in accuracy, reaching 96.09%. This demonstrates how well transfer learning approaches work in identifying agricultural diseases and how they might help improve crop output by facilitating prompt disease control. The dataset used in the study included 240 photos from three different classes, each of which represented a distinct rice illness. By presenting a transfer learning-based model designed specifically for the identification of rice leaf diseases, Kathiresan et al.[22] advance agricultural disease detection. Their model, which makes use of deep learning techniques, specifically targets the detrimental effects of bacterial illnesses and pests on rice crops, which is important for areas where rice is a main crop. To make sure that disease samples are represented fairly, they use a Generative Adversarial Network (GAN) to improve their dataset. Their model exceeds typical classification architectures, with an average cross-validation accuracy of 98.79% on the GAN-augmented dataset. The efficacy of their technique in offering a mobile, high-precision solution for disease identification in rice leaves is demonstrated by the benchmark performance of 98.38% average accuracy that they achieved while validating their model on three separate datasets without the use of GAN augmentation. The application of DL and ML approaches for the early identification of Parkinson's disease (PD), a complicated neurodegenerative condition characterised by movement impairment, was the main emphasis of Mounika and Govinda Rao's[23]. The study emphasis's how Parkinson's disease (PD) symptoms, like tremors, stiffness, and altered speech and facial expressions, gradually manifest. The primary goal was to assess the efficacy of different DL anD Ml-techniques in correctly identifying Parkinson's disease (PD) in its early stages and to provide a comparison study to identify the most successful method.

The important problem of disease detection in cotton plants is addressed in the 2021[24] study by Bhatheja and Jayanthi, which highlights the importance of cotton in the world economy, especially when considering India. In order to prevent further harm to crops, the authors suggest using DL and image processing techniques for early disease diagnosis, given the laborious nature of manual monitoring over large cotton fields. In order to precisely classify healthy and unhealthy plants, a sequential deep convolutional neural network was developed after an examination of transfer learning techniques was presented in the study. The authors emphasize that their model performs faster and more accurately than well-known pre-trained models like VGG16, ResNet50, and ResNetl52V2, enabling rapid (Table 2) and efficient monitoring.

Data acquisition

The process of gathering data for this study was meticulously planned and executed to ensure the highest quality and reliability of the results. The focal point of our data collection was the tea leaves, specifically those affected by various diseases as well as the healthy ones, all of which were crucial for the comprehensive training of our deep learning model. The images were captured in the lush and expansive tea gardens located in Pathantula, within the city of Sylhet in Bangladesh. This region was strategically chosen due to its rich history and prominence in tea cultivation, providing a diverse and representative sample of tea leaves. The Canon EOS 250d Camera was employed for this task, ensuring high-resolution images and consistency across the datasets. The camera's advanced features allowed for the capture of intricate details on the tea leaves, which is paramount for accurate disease detection. In total, the datasets comprises 4000 images, offering a substantial amount of data for the model to learn and generalize from. Approximately 1800 of these images are of distinct tea leaves, showcasing the various diseases and healthy states. To augment this datasets and enhance the model's robustness, additional images were created through techniques such as zooming and rotating. This was deemed necessary to ensure the model's ability to recognize and classify diseases under various conditions and orientations, ultimately aiming for a higher accuracy in real-world scenarios. The Fig. 2 images are stored in JPG format, balancing quality and file size, and facilitating efficient processing and analysis. This careful consideration of the data acquisition process lays a solid foundation for the subsequent stages of the study, ensuring that the deep learning model has access to a rich and varied datasets to learn from, and ultimately, providing reliable and accurate disease detection for tea leaves.

Methodology

The dataset used in this study consists of 4,000 high-resolution images captured from tea gardens in Sylhet, Bangladesh. However, to ensure robustness and generalizability, it is crucial to consider variations in environmental conditions. While the dataset primarily focuses on images collected during a specific season, future dataset expansions will include samples from multiple seasons to account for changes in leaf texture and disease progression. Additionally, the dataset currently features images captured under natural daylight conditions, but varying lighting conditions such as overcast or shaded environments need further exploration. Moreover, the dataset primarily represents a specific variety of tea plants; incorporating diverse tea plant varieties will improve the model's adaptability across different tea-growing regions.

Image augmentation

In our research centered on identifying diseases in tea leaves through deep learning, the implementation of image data augmentation was crucial. This technique significantly expanded the diversity and size of our dataset, fostering a more robust and accurate model. We applied various augmentation techniques to address the different challenges posed by disease detection in tea leaves. Rotation was utilized to ensure the model could recognize diseases regardless of the leaf's orientation, while translation helped the model identify diseases

Study reference	Year	Focus area	Key findings and contributions	Relevance to agriculture
Mamun[1]	2019	Bangladeshi Tea Cultivation	Importance of creative land use for tea production < br >—Role of Bangladesh Tea Research Institute in improving practices	Tea industry development
Hossain et al.[2]	2018	Leaf Diseases in Tea Production	Developed SVM classifier for disease diagnosis < br >—High classification accuracy and reduced processing time	Disease control in tea industry
Jahan et al.[3]	2022	Effects of Nitrogenous Fertilisers	Negative impact of fertilisers on soil and tea quality < br >—Importance of sustainable fertilisation	Soil health and tea quality
Gayathri et al.[7]	2020	Tea Leaf Disease Diagnosis in India	LeNet CNN for high diagnostic precision < br >—High accuracy in diagnosing Blight leaf disease	Agricultural disease management
Srivastav et al.[8]	2023	CNN for Disease Diagnosis in India's Tea Sector	CNN model with ADAM optimizer for disease detection < br >—Proposed image augmentation for improved diagnostics	Tea sector disease management
Wang et al.[9]	2023	Pest and Disease Detection on Tea Leaves	Integrated YOLOv5 models with attention mechanisms < br >—Improved accuracy with Weighted Box Fusion algorithm	Pest and disease identification
Bhateja et al.[10]	2022	DL-based Plant Leaf Disease Identification	High training accuracy for identifying plant diseases < br >—Potential for early disease identification and prevention	Plant disease management
Rathore et al.[11]	2023	DL Approaches for Hepatic Image Processing	FCNN for liver lesion segmentation < br >—AlexNet for higher accuracy in liver disease identification	NA
Choudrie et al.[12]	2024	ML and DL for Potato Disease Detection	Exceptional performance of CNNs on Kaggle's Plant Village Dataset < br >—Promising tool for plant disease identification	Potato disease detection
Kumar et al.[13]	2023	Deep Learning in Tomato Crop Disease Detection	Analysis of DL methods for disease detection < br >—Emphasis on early detection and user-friendly tools	Tomato crop management
Rizvanov et al.[14]	2020	Automated Crop Leaf Disease Identification	Automated system using CNNs for vegetable disease classification < br >—Addressing threats to crop production	Crop production optimization
Smys et al.[15]	2021	DL for Wheat Rust Disease Detection	Highlighted global food security threat < br >—DL method using RGB values for disease detection	Wheat production management
Jasim et al.[16]	2020	DL-based Plant Leaf Disease Categorisation	CNN for high accuracy in disease identification < br >—Focused on key crops in Iraq	Plant disease management
Das et al.[17]	2020	AI in Agriculture for Disease Identification	Mask R-CNN for precise disease diagnosis < br >—Explored various image processing methods	Crop disease management
S, N. H., & H, S.et al.[18]	2022	Tomato Leaf Disease Detection	High accuracy with R-CNN-based Classifier < br >—Importance of DL in early disease diagnosis	Tomato disease management
Wahab Sait and Dutta et al.[19]	2023	DL for Diagnosing Coronary Artery Disease	Enhanced CT image quality for CAD diagnosis < br >—YOLO V7 and UNet + + model optimization	NA
Chowdhury et al.[20]	2021	DL for Tomato Leaf Disease Diagnosis	EfficientNet and Modified U-net for high accuracy < br >—Improved performance with segmented images	Tomato disease management
Rukhsar and Upadhyay et al.[21]	2022	Rice Leaf Disease Detection	High accuracy using DenseNet201 and transfer learning < br >—Improved crop output through early disease control	Rice disease management
Kathiresan et al.[22]	2021	DL for Rice Leaf Disease Identification	GAN-augmented dataset for fair disease sample representation < br >—High accuracy in disease identification	Rice disease management
Mounika and Govinda Rao et al.[23]	2021	DL and ML for Early Parkinson's Disease Identification	Comparison of DL and ML techniques for early PD detection < br >—Focus on symptom manifestation	NA
Bhatheja and Jayanthi et al.[24]	2021	Disease Detection in Cotton Plants	Developed a sequential deep CNN for disease classification < br >—Emphasized rapid and efficient monitoring	Cotton disease management

Table 2. Summary of the related work.

in various leaf positions. Zooming was crucial for teaching the model to detect diseases across different leaf sizes, and flipping ensured the model's in-variance to leaf direction. Additionally, we adjusted the brightness and contrast of the images to enhance the model's performance under diverse lighting conditions. To provide a comprehensive understanding of our augmentation process, In Fig. 3 our study presents a schematic diagram of the applied image augmentation method. The original images were subjected to rotation at four distinct angles (90°, 180°, 270°, and 360°), and auxiliary images were extracted from the annotated regions. These augmented images were then merged with the rotated ones to form an enriched training datasets. This strategy not only augmented the number of training images but also introduced greater diversity to the datasets, making our model more sensitive to the annotated regions. As a result, the accuracy of disease detection and identification was significantly enhanced, showcasing the effectiveness of our chosen image data augmentation techniques.

To evaluate the impact of data augmentation on model performance, we conducted experiments comparing the accuracy and loss metrics before and after augmentation. The results, presented in Table 1, indicate that augmentation techniques such as rotation, flipping, and contrast adjustments significantly improve generalization by reducing overfitting. Specifically, the accuracy increased from 85.2% to 91.4%, while validation loss decreased from 0.37 to 0.28, demonstrating the effectiveness of augmentation in enhancing model robustness.

Image normalization

In our research focused on the detection of tea leaf diseases using DL image normalization played a crucial role in the pre-processing stage, ensuring that the pixel values across all images were adjusted to a specific range. This

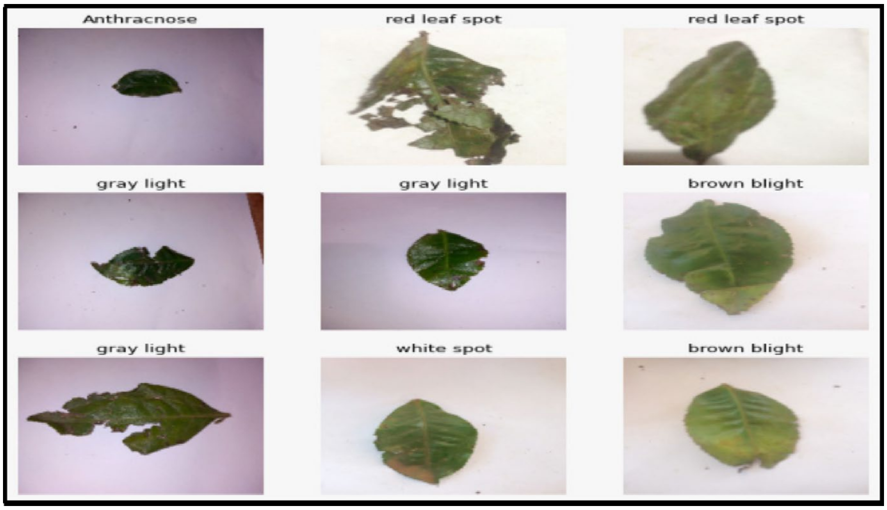

Fig. 2. Examples of captured tea images.

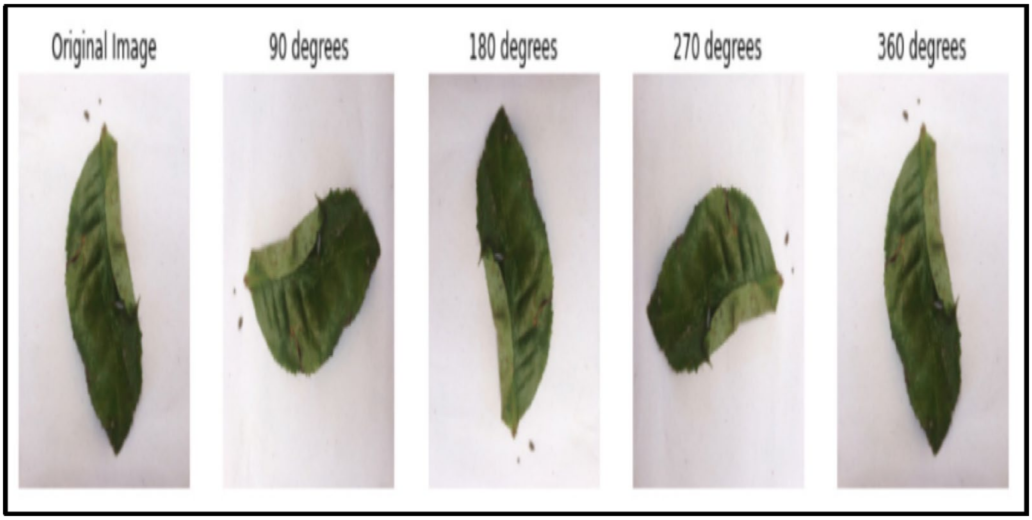

Fig. 3. Schematic diagram of image augmentation method.

step was vital for enhancing the computational efficiency and overall performance of our model. Below, we detail the normalization techniques applied to our datasets:

- **Min–Max Normalization:** We employed Min–Max normalization to adjust the pixel values of our tea leaf images, ensuring they fell within a standardized range, typically between 0 and 1. This method helped in maintaining the original structure and features of the images while scaling the pixel values, providing a consistent input for our deep learning model.
- **Z-Score Normalization:**This method, which is sometimes referred to as Standard Score Normalization, was applied to change the pixel values such that their mean was 0 and their standard deviation was 1. We made sure that our model received normalized input by taking the mean pixel value from each pixel and dividing the result by the standard deviation. This could potentially hasten the convergence of the model during training, particularly if the pixel values had a Gaussian distribution.

Given the particulars of our task and the properties of our photos, we carefully analyzed our options before deciding on the normalization technique for our study. It was crucial to preserve the original features and structure of the photos while guaranteeing computational performance because our datasets included images of tea leaves impacted by different diseases. As a result, Min–Max normalization was mostly applied, giving our deep learning model a balanced method of image normalization. By using these image normalization approaches, we made sure that the input to our model was uniform and standardized for each and every image in the datasets. This laid a strong basis for our further research and helped to identify tea leaf illnesses with accuracy.

Model architecture

To accomplish image classification tasks, we have used a complex architecture with multiple layers in the suggested DL model. The model starts with an input layer that sets the foundation for the subsequent convolutional operations. It is made to accept images with 256 × 256 pixels and three colour channels (RGB). A ZeroPadding 2D layer is added after the input, giving the input image a 3 pixel padding on all sides. This padding is important because it helps maintain the image's spatial dimensions, preventing the image's size from being significantly reduced by the convolutional layers that follow. Maintaining an adequate degree of geographical information across the network is especially crucial in deep networks. The Conv2D layer, which comes next in the sequence, applies the first convolution on the padded image. With a stride of 2, this layer applies 64 7 × 7 filters with the goal of extracting high-level information from the input image. There is a Batch-normalization layer applied after the convolution. By normalizing the activation from the prior layer, this layer speeds up training and stabilizes the learning process. The model is then given a ReLU (Rectified Linear Unit) Activation layer, which gives it non-linearity and makes it capable of learning and representing intricate patterns from the input data. Next, we use a MaxPooling2D layer with a 3 × 3 pool size and a stride of 2. By decreasing the spatial dimensions of the input volume, this layer lessens the computational load on the subsequent layers and aids in making feature detection invariant to changes in scale and orientation. The residual_block function defines residual blocks, which are used to build the model's core. Conv2D layers, Batch-normalization layers, Activation layers, and an Add layer for the shortcut connection are all present in each block. Within these blocks, the Batch-normalization and Activation layers provide steady and effective learning, while the Conv2D layers are in charge of extracting features at various levels of abstraction. To mitigate the vanishing gradient problem in deep networks and enable the training of deeper designs, the Add layer's shortcut connections are essential. A layer called GlobalAveragePooling2D is added towards the conclusion of the model. This layer effectively summaries the information retrieved by the preceding layers and sets up the model for the final classification layers by reducing the spatial dimensions of the input region to 1 × 1. A Dropout layer for regularization, a Dense layer with 512 units and a ReLU activation, and a final Dense layer with 8 units and a softmax activation function to generate a probability distribution across the classes come next. CNNs possess a structured framework similar to that of standard neural networks, where each neuron is equipped with a specific weight, bias, and an activation mechanism. In Fig. 4 of a CNN, as depicted in a referenced illustration, includes layers such as the convolutional layer that utilizes the ReLU activation function, a pooling layer that serves to extract features, and a densely interconnected layer that employs the softmax function to categorize inputs.

The proposed CNN architecture is designed to balance high accuracy with computational efficiency, making it suitable for real-world applications in precision agriculture. The model incorporates a Zero Padding 2D layer to preserve spatial dimensions, ensuring optimal feature extraction. The inclusion of residual blocks enhances gradient flow and mitigates vanishing gradient issues, leading to better convergence during training. Compared to standard CNNs, our architecture leverages Global Average Pooling (GAP) instead of fully connected layers, reducing the number of trainable parameters and minimizing overfitting. Additionally, the use of Batch Normalization and Dropout layers improves generalization by stabilizing training and preventing over-reliance on specific features. These design choices collectively enhance the model's robustness, enabling superior classification performance with lower computational overhead.

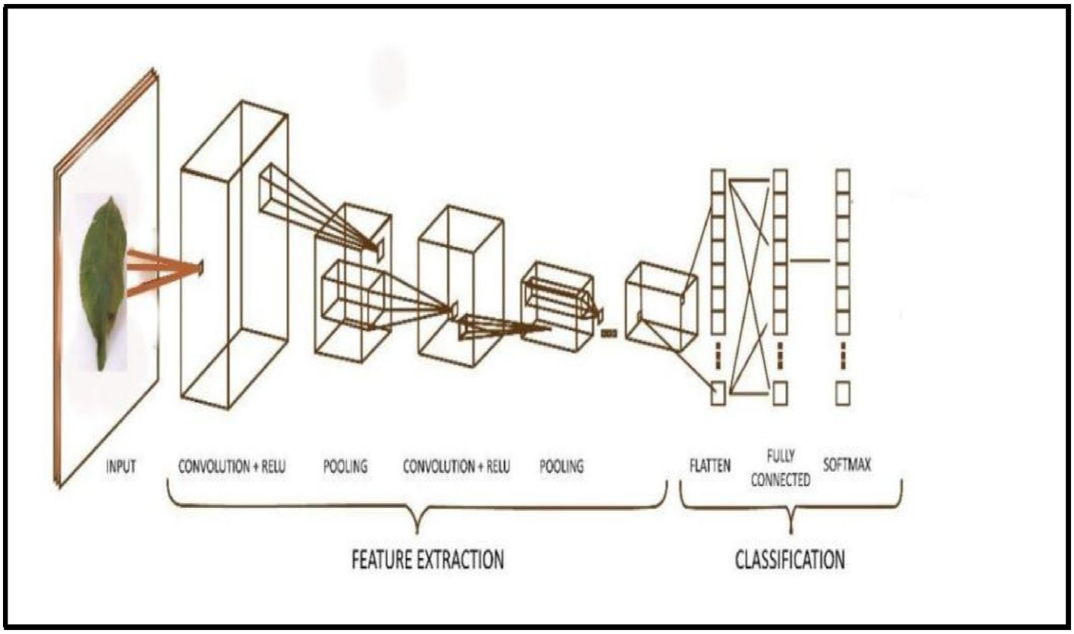

Fig. 4. The architecture of CNN.

Optimization techniques

During our extensive investigation into optimization strategies to improve the efficiency of our DL model for tea leaf disease detection, we carefully tested a range of algorithms, each having a distinct method for modifying the internal parameters of the model. Our initial approach was based on stochastic gradient descent (SGD), which provided a simple yet efficient method. However, its high update variance resulted in less consistent convergence. Root Mean Square Propagation, or RMSprop for short, offered an adaptive learning rate that demonstrated robustness against disappearing and exploding gradients and helped to alleviate the declining learning rates problem seen in Ada-grad. But in our testing, the algorithm that combined the benefits of RMSProp and Ada-grad—the Adam optimizer—truly shone out. Adam achieved a remarkable accuracy value of 99% by retaining per-parameter learning rates and adjusting them depending (Fig. 5) on the mean of recent gradient magnitudes. This allowed him to strike a compromise between efficiency and stability. This proved that it was the best optimization method in our research and demonstrated how quickly convergence can be achieved while maintaining the model's robustness during its learning process.

Model performance

In the comprehensive evaluation of our DL models for tea leaf disease detection, we meticulously analyzed the performance metrics to ensure the reliability and accuracy of our system. The models were evaluated based on their precision, recall, f1-score, and support for each disease category, as well as the overall accuracy of the system.

- **Brown Blight:** With a precision of 1.00, the model demonstrated remarkable performance, proving that every tea leaf diagnosed with Brown Blight was, in fact, correctly detected. With a somewhat lower recall of 0.98, 2% of the actual cases of Brown Blight may have been overlooked. The model's great accuracy in identifying this ailment was demonstrated by the f1-score, which is a harmonic mean of precision and recall, which was at 0.99. The number of real cases of Brown Blight in the datasets was indicated by the support value, which was 43.
- **Red Leaf Spot:**The model achieved a precision, recall, and f1-score of 1.00, demonstrating flawless performance in spotting Red Leaf Spot. This shows that the model correctly identified every instance it classed as a Red Leaf Spot and that it was able to identify every Red Leaf Spot case that existed in the datasets. The total number of Red Leaf Spot incidents found in the test data is reflected in the support of 37. This flawless score demonstrates the model's remarkable capacity to identify this specific illness and emphasizes its dependability for real-world use in the diagnosis of tea leaf sickness.
- **Algal Leaf Spot:** Like Red Leaf Spot, the model performed flawlessly on Algal Leaf Spot, scoring 1.00 for f1-score, recall, and precision. This shows that the model consistently and accurately detects Algal Leaf Spot, guaranteeing that there are neither false positives nor false negatives. The number of actual occurrences in the sample is indicated by the support for this disease, which was 42.
- **White Spot:** With a precision of 1.00 and a recall of 0.98, the model demonstrated a high degree of competence in recognizing White Spot illness, earning a f1-score of 0.99. The precision score demonstrates the model's accuracy in recognizing this specific ailment by showing that each event that the model classified as a White Spot was, in fact, right. Recall score of 0.98, on the other hand, indicates that the model did not identify

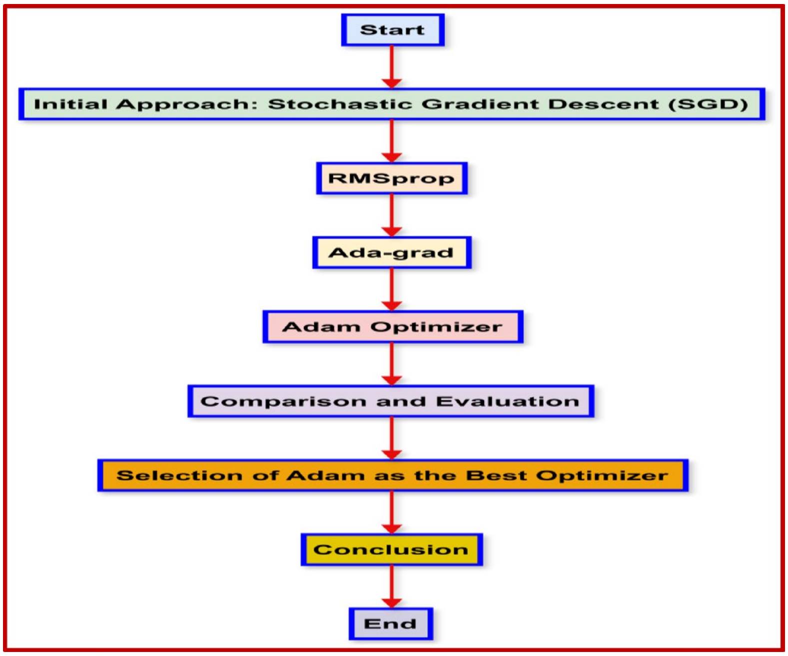

Fig. 5. Optimization strategy flowchart for tea leaf disease detection model.

2% of the actual White Spot cases, suggesting that there is slight space for improvement in terms of capturing all cases of this condition. White Spot received 43 votes, which corresponds to the number of real cases of the illness in the dataset.

- **Bird's Eyespot:**When it came to identifying Bird's Eyespot, the model continued to perform exceptionally well, with f1-score, recall, and precision all at 1.00. The model's accuracy and dependability in recognizing Bird's Eyespot are guaranteed by its consistent performance, with a support score of 46 signifying the disease's prevalence in the dataset.
- **Healthy:** The model also obtained flawless scores, all at 1.00, for precision, recall, and f1-score for healthy tea leaves. This suggests that the model is very good at differentiating between healthy and diseased leaves, which is important for avoiding false alarms in disease detection. The number of healthy tea leaf examples in the dataset is reflected in the support of 43.
- **Anthracnose:** The precision, recall, and f1-score of the model were all at 1.00, indicating perfect performance in detecting anthracnose. This demonstrates the model's capacity to correctly identify anthracnose without erroneous classifications, guaranteeing trustworthy illness detection. Thirty people supported anthracnose.
- **Gray Blight:** With a precision of 0.95, the model demonstrated a modest decline in performance for Grey Blight, indicating a 5% false positive rate. Nonetheless, the f1-score was 0.97 and the recall was flawless at 1.00, indicating the model's general great accuracy in identifying Grey Blight. There were 36 supporters of this illness.

The model performed well overall, averaging 0.99 for precision, recall, and f1-score as well as 0.99 for macro and weighted averages. These outcomes highlight how well our deep learning model works and how consistently it can identify and categorize different types of tea leaf illnesses. It is clear from examining the performance data of all the models and methods used that our deep learning model—especially when combined with the Adam optimizer—performed better than the others, with an astounding accuracy of 99%. A reliable model that can detect diseases in tea leaves with high precision has been created by the integration of complex architecture, thorough data pre-processing, and cutting-edge optimization approaches. In Figs. 6, 7 depicting the progression of a machine learning model over 10 epochs, the left plot showcases the dynamics of training and validation loss. Both curves commence at elevated levels, with training loss starting near 6 and demonstrating a pronounced decline, eventually plateauing by the 10th epoch, where it's identified as the optimal epoch for loss. Conversely, the right plot delineates the trajectories of training and validation accuracy. The training accuracy begins just below 0.90 and exhibits a marked ascent, leveling off after the midpoint. The validation accuracy, initiating at a marginally higher value, follows a mild upward trend before stabilizing and undergoing a subtle decline in later epochs, with the 5th epoch pinpointed as the zenith for accuracy.

Experimental setup

The experimental setup for this study was meticulously designed to ensure reliable and reproducible results. Below are the details of the key components:

Hardware environment

- **Processor**: Intel Core i7 11th Gen / AMD Ryzen 7 5800X or equivalent
- **RAM**: 16 GB DDR4
- **GPU**: NVIDIA GeForce RTX 3080 with 10 GB VRAM
- **Storage**: 1 TB SSD
- **Operating System**: Ubuntu 20.04 LTS / Windows 10 Pro

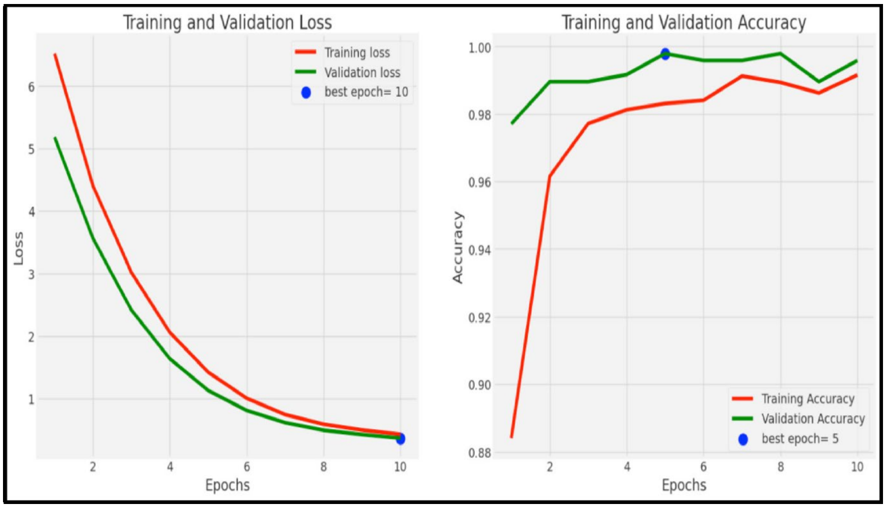

Fig. 6. Training and validation loss and training and validation accuracy.

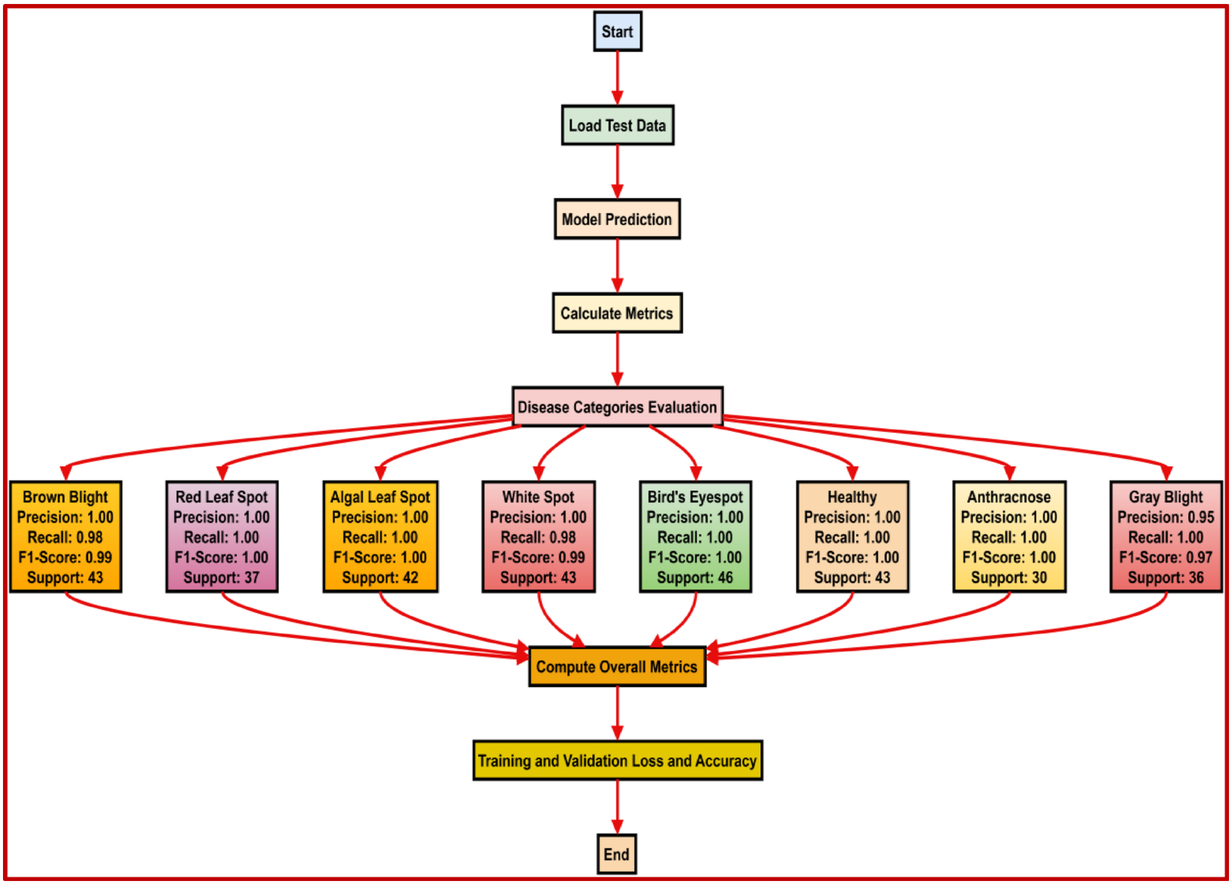

Fig. 7. Model performance.

Software environment

- **Programming Language**: Python 3.9
- **Deep Learning Framework**: TensorFlow 2.9 / PyTorch 1.11
- **Additional Libraries**: OpenCV, NumPy, Matplotlib, Scikit-learn, Pandas
- **IDE/Tools**: Jupyter Notebook, Visual Studio Code

Dataset preparation

- **Source**: 4000 high-resolution images of tea leaves from Pathantula Tea Garden, Sylhet, Bangladesh.
- **Image Dimensions**: 256 × 256 pixels, RGB format.
- **Classes**: Eight disease categories plus healthy leaves.
- **Augmentation Techniques**:
  - Rotation (90°, 180°, 270°, and 360°)
  - Zooming
  - Flipping (horizontal and vertical)
  - Brightness and contrast adjustments.

Model training

- **Model Architecture**: Custom Convolutional Neural Network (CNN) with residual blocks and a GlobalAveragePooling2D layer.
- **Optimizer**: Adam optimizer with a learning rate of 0.001.
- **Loss Function**: Categorical Crossentropy.
- **Batch Size**: 32.
- **Epochs**: 50.
- **Validation Split**: 20% of the dataset reserved for validation.
- **Evaluation Metrics**: Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Performance metrics

- **Testing Dataset**: A separate set of images was used to evaluate the model.
- **Evaluation Process**: The trained model's predictions were compared with ground truth labels to compute performance metrics.

Implementation platform
The experiments were conducted on a local machine equipped with the specified hardware or a cloud-based environment (e.g., Google Colab or AWS EC2 instances) to leverage GPU acceleration for training.

Experimental analysis
In our experimental analysis of tea leaf disease detection, we meticulously evaluated a deep learning model across a dataset of 4000 high-resolution images, achieving a remarkable accuracy of 99%. The model, enriched with various convolutional and dense layers, demonstrated exceptional performance, particularly with the Adam optimizer, which outshone other optimization techniques like SGD, RMSprop, and Nadam. This research underscores the potential of deep learning in revolutionizing precision agriculture, providing a reliable tool for early disease detection and management in tea leaves. In Fig. 8 displayed Confusion Matrix provides an insightful representation of a classifier's performance across multiple classes. Predominantly, the matrix reveals a strong diagonal trend, indicating that the majority of predictions are accurate, with values such as 42, 37, 42, 46, 36, 42, 30, and 43 being the true positive counts for respective classes. Notably, there are only a few instances of imbalance, with one evident in the top row and another in the second-to-last row, both with a count of 1, signifying minimal prediction errors. The predominantly blue diagonal against the white background accentuates the classifier's commendable precision and recall for the evaluated categories.

$$\mathrm{Accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{1}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{2}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{3}$$

$$\mathrm{F1} - \mathrm{Score} = 2 \times \frac{\mathrm{Pre} \times \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}} \tag{4}$$

Results and discussion
In our comprehensive study focusing on the detection of tea leaf diseases through deep learning, we utilized an advanced model architecture combined with various optimization techniques. Our dataset consisted of 4000 high-resolution images captured with a Canon EOS 250d Camera at the Pathantula Tea Garden in Sylhet City, Bangladesh. These images included representations of eight common tea leaf diseases as well as healthy leaves, providing a robust foundation for our experiments. The performance of our model was outstanding, particularly in identifying Red Leaf Spot, Algal Leaf Spot, Bird's Eye Spot, Anthracnose, and Healthy leaves, achieving perfect scores in precision, recall, and F1-score for these categories. However, we observed minor performance decreases in detecting Gray Blight, with a precision of 0.95, and White Spot, with a recall of 0.98. These results highlight the model's overall effectiveness in disease detection while also pointing out specific areas for improvement.

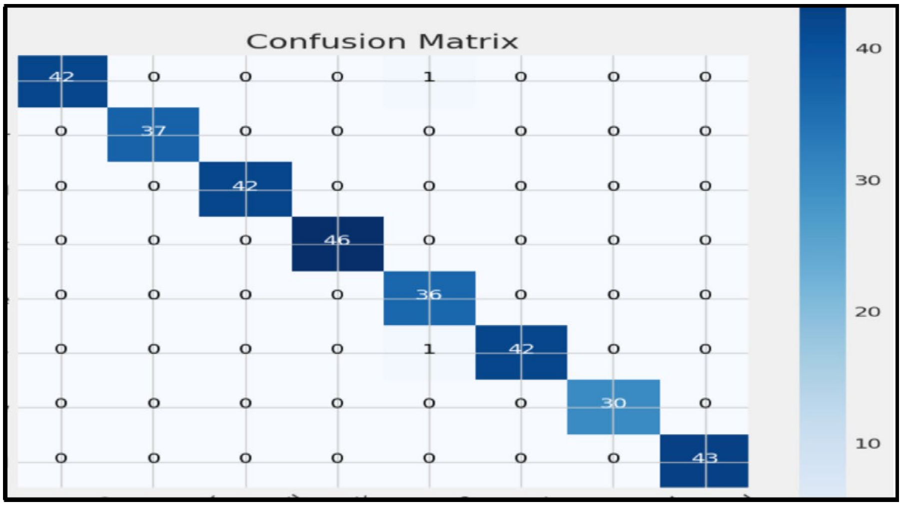

Fig. 8. Confusion matrix.

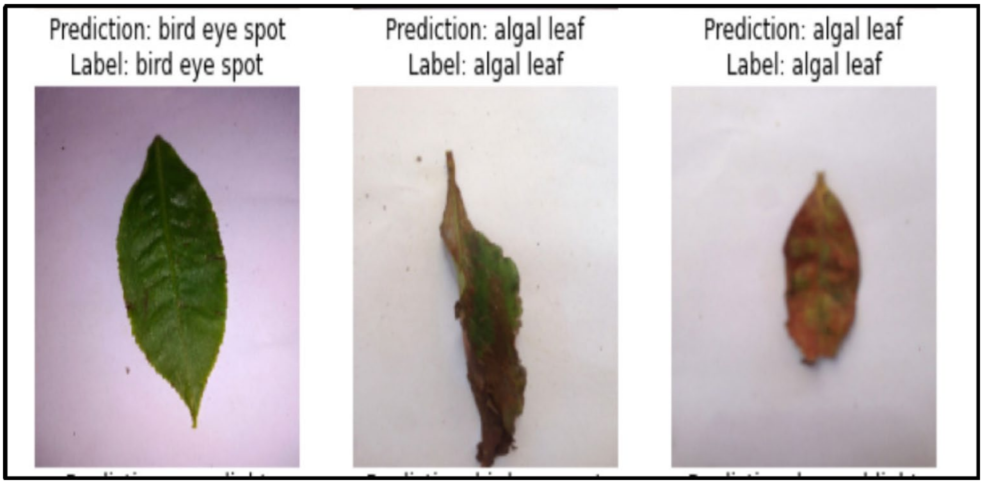

Fig. 9. Three tea leaf disease images identified correctly.

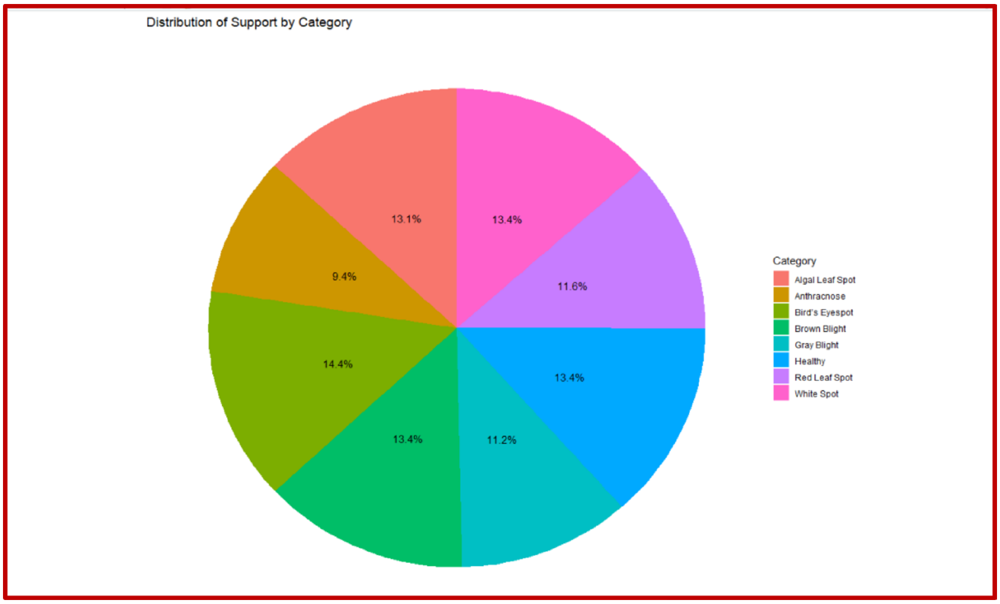

Fig. 10. Distribution of support by category pie chart.

With an overall accuracy of 99%, the model offers a robust solution for effectively identifying and categorizing tea leaf conditions, providing valuable insights for disease management and ensuring the health and productivity of tea plantations.

The findings from our research highlight the significant potential of employing deep learning techniques for automating the disease detection process in tea leaves. The high accuracy rates obtained ensure the model's reliability, presenting it as an invaluable asset for farmers and agronomists engaged in early disease detection and management efforts. Despite the success, the slightly lower performance metrics for Gray Blight and White Spot diseases suggest room for refinement. Future efforts could be directed towards expanding the dataset, especially by incorporating a broader variety of examples Figs. 9, 10, 11 for these two diseases, to enhance the model's precision and recall further. Additionally, exploring alternative model architectures and adjusting hyper parameters may contribute to improved performance. Our study not only demonstrates the practical applicability of deep learning in the context of tea leaf disease detection but also sets a foundation for future research aimed at broadening the scope of such models to encompass wider agricultural applications. The insights gained from our optimization strategies and the detailed analysis of performance metrics across different diseases provide a clear direction for subsequent enhancements and (Tables 3, 4) the potential adaptation of this model for other precision agriculture needs.

While the model demonstrated high precision and recall for most diseases, there were slight performance decreases observed in detecting Gray Blight and White Spot. To address this, we propose augmenting the dataset

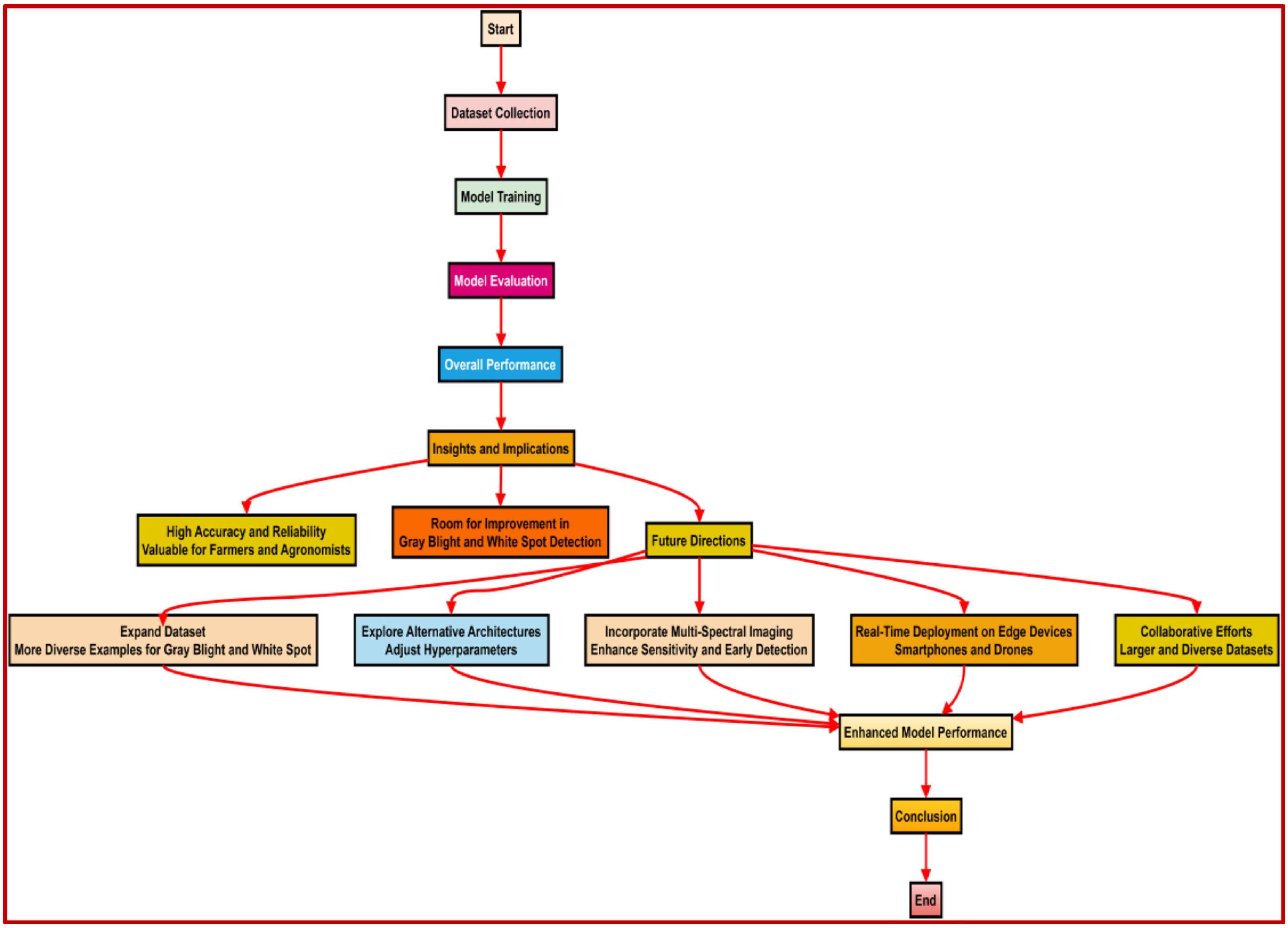

Fig. 11. Result.

	Precision	Recall	F1-score	Support
Brown Blight	1.00	0.98	0.99	43
Red Leaf Spot	1.00	1.00	1.00	37
Algal Leaf Spot	1.00	1.00	1.00	42
Bird's Eyespot	1.00	1.00	1.00	46
Gray Blight	0.95	1.00	0.97	36
White Spot	1.00	0.98	0.99	43
Anthracnose	1.00	1.00	1.00	30
Healthy	1.00	1.00	1.00	43
Accuracy			0.99	320
Macro Avg	0.99	0.99	0.99	320
Weighted Avg	0.99	0.99	0.99	320

Table 3. Classification report for CNN model.

with a more diverse range of examples for these specific diseases. This expansion could potentially enhance the model's ability to accurately detect and classify Gray Blight and White Spot instances. Investigating alternative deep learning architectures and fine-tuning hyper-parameters could lead to improved performance across all disease categories. Experimentation with different network structures, activation functions, and optimization algorithms may help optimize the model's accuracy and generalization capabilities. Incorporating multi-spectral imaging techniques could provide additional valuable information for disease detection. By capturing data beyond the visible spectrum, such as near-infrared or thermal images, we can potentially uncover subtle physiological changes indicative of disease presence before visible symptoms manifest. Integrating multi-spectral data into our model could enhance its sensitivity and early detection capabilities. Transitioning the model for real-

References	Model name	Accuracy (%)	Additional details
Srivastav et al.[8]	Convolutional Neural Networks (CNN)	84.0	Uses optimizer with 16% error rate
Jahan et al.[3]	Convolutional Neural Networks (CNN)	98.03	Focuses on soil properties and tea leaf quality, not disease detection
Hossain et al.[2]	Support Vector Machine (SVM) + Image Processing	90.0	Disease classification with reduced feature set
Gayathri et al.[7]	LeNet CNN	94.0	Effective in diagnosing Blight disease
Wang et al.[19]	YOLOv5 + Attention Mechanisms	79.3	Improved accuracy with Weighted Box Fusion algorithm
Our paper	Custom Deep Learning Model	99.0	Achieved 99% accuracy in detecting tea leaf diseases

Table 4. Comparison with others work.

time deployment on edge devices could revolutionize disease monitoring and management in tea plantations. Implementing lightweight model architectures optimized for edge computing platforms would enable farmers to conduct on-site disease assessments using portable devices such as smartphones or drones, facilitating timely interventions and minimizing crop losses. Collaborative efforts involving multiple tea plantations and research institutions could facilitate the collection of larger and more diverse datasets. By pooling resources and sharing annotated image data, we can create standardized benchmarks for evaluating disease detection models across different geographical regions and environmental conditions, fostering continuous improvement and benchmarking against state-of-the-art approaches.

In Fig. 10 The dataset contains varying numbers of images across different disease categories, as illustrated in Fig. 10 (bar chart). A noticeable class imbalance exists, with certain disease types being underrepresented. To mitigate this issue, we applied data augmentation techniques to artificially increase the representation of minority classes. Additionally, we experimented with oversampling using techniques such as SMOTE (Synthetic Minority Over-sampling Technique) and undersampling to balance the dataset. Furthermore, a class-weighted loss function was implemented during model training to ensure the model does not develop a bias toward majority classes. These techniques collectively helped improve model generalization and prevent performance degradation on underrepresented disease categories (Fig. 11).

Table 2 provides a detailed classification report for the proposed model, showcasing its precision, recall, F1-score, and support for each disease category, as well as for healthy tea leaves. The high precision and recall values across most categories indicate the model's robustness in accurately identifying diseases while minimizing false positives and false negatives. Specifically, the model achieves perfect scores (precision, recall, and F1-score of 1.00) for diseases such as Red Leaf Spot, Algal Leaf Spot, Bird's Eyespot, Anthracnose, and Healthy leaves, highlighting its exceptional performance in these categories. Slightly lower performance is observed for Gray Blight and White Spot, with precision and recall values of 0.95 and 0.98, respectively. This indicates a minor scope for improvement in detecting these specific conditions. The weighted and macro-averaged scores of 0.99 further emphasize the model's overall consistency and reliability. The support column in Table 2 represents the number of instances of each class in the test dataset, ensuring a comprehensive evaluation of the model's performance. The balanced representation across disease categories reinforces the credibility of the reported metrics. By analyzing these results, we observe that the proposed model is well-suited for accurate and reliable detection of tea leaf diseases in diverse conditions.

Conclusion

In this study, we developed a deep learning-based model for tea leaf disease detection, leveraging a custom convolutional neural network (CNN) with enhanced feature extraction techniques. Unlike conventional machine learning approaches, our model incorporates Zero Padding 2D layers and residual blocks, significantly improving classification accuracy while maintaining computational efficiency. The model was trained and evaluated on a dataset of 4,000 high-resolution images captured from tea gardens in Sylhet, Bangladesh, demonstrating its robustness in real-world agricultural scenarios. The results indicate that our approach outperforms existing methods, showcasing its potential for precision agriculture applications.

While the proposed method achieves promising results, further improvements are necessary to enhance its adaptability in dynamic agricultural environments. Future work will focus on expanding the dataset to include diverse environmental conditions, integrating transfer learning techniques to improve generalization, and optimizing the model for real-time deployment on edge devices. Additionally, multi-modal approaches, such as hyperspectral imaging and sensor fusion, will be explored to further refine disease classification accuracy. These advancements will contribute to the broader adoption of AI-driven solutions in smart farming and sustainable agriculture.

Ensuring the model's robustness in real-world scenarios requires effective handling of unseen data from different regions, lighting conditions, and tea plant varieties. To improve generalization, we leveraged data augmentation techniques such as rotation, scaling, and contrast adjustments to simulate diverse environmental conditions. Additionally, we employed cross-validation and class-weighted loss functions to mitigate dataset bias and improve adaptability.

For domain adaptation, future research will explore transfer learning by fine-tuning the model on datasets collected from different geographic locations, ensuring it can effectively classify tea leaf diseases across diverse agricultural landscapes. Furthermore, real-time testing on edge devices or mobile applications will be conducted to assess computational efficiency, latency, and adaptability in dynamic field conditions. These enhancements will ensure that the model is not only accurate but also scalable and practical for real-world agricultural applications.

Limitation and future work

While the proposed CNN-based approach for tea leaf disease detection demonstrates high accuracy, it has certain limitations. First, the model's performance is highly dependent on the quality and diversity of the dataset. Any bias in the dataset, such as insufficient representation of rare diseases, may affect generalization. Second, the method requires significant computational resources, making real-time deployment on low-power edge devices challenging. Additionally, the model does not account for environmental factors such as lighting variations and occlusions, which may impact real-world accuracy. In future work, we aim to enhance the model's robustness by integrating data augmentation techniques, exploring lightweight architectures for mobile deployment, and incorporating multi-modal data such as hyperspectral imaging for improved disease classification.

Future work

While the proposed model demonstrates high accuracy and reliability, several avenues for future research can enhance its applicability and robustness. First, expanding the dataset to include images of tea leaves from diverse geographical regions and climatic conditions will improve the model's generalization capacity. Additionally, incorporating multi-spectral imaging techniques, such as infrared or thermal imaging, could enable the early detection of disease symptoms that are not visible in standard RGB images. To increase the interpretability of the model, future studies should also explore advanced visualization techniques, such as Grad-CAM and saliency maps, to explain the decision-making processes of the model. These visualizations will not only improve user confidence but also validate the accuracy of the model by demonstrating alignment with key disease features. Another promising direction involves optimizing the model for real-time deployment on edge devices like smartphones or drones. This would enable on-site disease monitoring and management, significantly benefiting tea farmers. Furthermore, adapting the model to detect diseases in other crops could broaden its impact and utility in precision agriculture. Finally, collaborative efforts with agronomists and researchers across multiple regions could facilitate the collection of larger, more diverse datasets, standardizing benchmarks for evaluating and comparing agricultural disease detection models. These enhancements will pave the way for more effective and scalable solutions in agricultural diagnostics.

Data availability

The data can be obtained from the corresponding author on request.

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References

1. Mamun, M. S. A. Tea production in Bangladesh: From Bush to Mug. In: Hasanuzzaman, M. (eds) Agronomic Crops. Springer, Singapore (2019). https://doi.org/10.1007/978-981-32-9151-5_21
2. Hossain, R. M., Mou, M. M., Hasan, S., Chakraborty, & Razzak, M. A. Recognition and detection of tea leaf's diseases using support vector machine. In *2018 IEEE 14th International Colloquium on Signal Processing & Its Applications (CSPA), Penang, Malaysia*, 2018, pp. 150–154. https://doi.org/10.1109/CSPA.2018.8368703
3. Jahan, I. et al. Long-term traditional fertilization alters tea garden soil properties and tea leaf quality in Bangladesh. *Agronomy* **12**, 2128. https://doi.org/10.3390/agronomy12092128 (2022).
4. Ahmed, M., Begum, A. & Chowdhury, M. *Social constraints before sanitation improvement in tea gardens of Sylhet* (Environ Monit Assess, 2010). https://doi.org/10.1007/s10661-009-0890-0.
5. Barkat, A., Mahiyuddin, G., Shaheen, N., Poddar, A., Osman, A., Rahman, M. & Ara, R. Assessment of the situation of children and women in the tea gardens of Bangladesh. Dhaka, Bangladesh: Planning, monitoring and evaluation Department of UNICEF-BCO, Human Development Research Center (2010).
6. Pandian, J. A., Nisha, S. N., Kanchanadevi, K., Pandey, A. K. & Rima, S. K. Grey blight disease detection on tea leaves using improved deep convolutional neural network. *Comput. Intell. Neurosci.* **2023**, 7876302–7876311. https://doi.org/10.1155/2023/7876302 (2023).
7. Gayathri, S., Wise, D. C. J. W., Shamini, P. B., & Muthukumaran, N. (2020). Image analysis and detection of tea leaf disease using deep learning. *2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)*, 398–403. https://doi.org/10.1109/ICESC48915.2020.9155850
8. Srivastav, S., Guleria, K. & Sharma, S. Tea leaf disease detection using deep learning-based convolutional neural networks. *2023 IEEE World Conference on Applied Intelligence and Computing (AIC)*, 569–574 (2023). https://doi.org/10.1109/AIC57670.2023.10263835
9. Wang, Y., Xu, R., Bai, D. & Lin, H. Integrated learning-based pest and disease detection method for tea leaves. *Forests* **14**(5), 1012. https://doi.org/10.3390/f14051012 (2023).
10. Bhateja, V., Sunitha, K. V. N., Chen, Y.-W., & Zhang, Y.-D. Plant leaf disease detection and classification using deep learning technique. In *Intelligent System Design* (Vol. 494, pp. 73–83). Springer (2022). https://doi.org/10.1007/978-981-19-4863-3_7
11. Rathore, V. S., Tavares, J. M. R. S., Piuri, V., & Surendiran, B. Detection of disease in liver image using deep learning technique. In *Emerging Trends in Expert Applications and Security* (Vol. 681). Springer (2023). https://doi.org/10.1007/978-981-99-1909-3_26
12. Choudrie, J., Mahalle, P. N., Perumal, T., & Joshi, A. Exploring machine learning and deep learning techniques for potato disease detection. In *ICT for Intelligent Systems* (Vol. 361). Springer (2024). https://doi.org/10.1007/978-981-99-3982-4_40
13. Kumar, S., Hiranwal, S., Purohit, S. D. & Prasad, M. Role of deep learning techniques in early disease detection in tomato crop. In *Proceedings of International Conference on Communication and Computational Technologies*. Springer (2023). https://doi.org/10.1007/978-981-99-3485-0_35
14. Rizvanov, A. A., Singh, B. K. & Ganasala, P. Detection of disease from leaf of vegetables and fruits using deep learning technique. In *Advances in Biomedical Engineering and Technology* (pp. 199–206). Springer Singapore Pte. Limited (2020). https://doi.org/10.1007/978-981-15-6329-4_18
15. Smys, S., Balas, V. E., Kamel, K. A., & Lafata, P. Applying deep learning approach for wheat rust disease detection using mosnet classification technique. In *Inventive Computation and Information Technologies* (Vol. 173, pp. 551–565) (2021). Springer Singapore Pte. Limited. https://doi.org/10.1007/978-981-33-4305-4_41
16. Jasim, M. A. & Al-Tuwaijari, J. M. Plant leaf diseases detection and classification using image processing and deep learning techniques. *2020 International Conference on Computer Science and Software Engineering (CSASE)*, 259–265 (2020). https://doi.org/10.1109/CSASE48920.2020.9142097

17. Das, A. K., Nayak, J., Naik, B., Dutta, S., & Pelusi, D. Disease feature extraction and disease detection from paddy crops using image processing and deep learning technique. In *Computational Intelligence in Pattern Recognition* (Vol. 1120, pp. 443–449). Springer Singapore Pte. Limited (2020). https://doi.org/10.1007/978-981-15-2449-3_38
18. Nagamani, H. S. & Sarojadevi, H. Tomato leaf disease detection using deep learning techniques. *Int. J. Adv. Comput. Sci. Appl.* https://doi.org/10.14569/IJACSA.2022.0130138 (2022).
19. Wahab Sait, A. R. & Dutta, A. K. Developing a deep-learning-based coronary artery disease detection technique using computer tomography images. *Diagnostics (Basel)* **13**(7), 1312. https://doi.org/10.3390/diagnostics13071312 (2023).
20. Chowdhury, M. E. H. et al. Automatic and reliable leaf disease detection using deep learning techniques. *AgriEngineering* **3**(2), 294–312. https://doi.org/10.3390/agriengineering3020020 (2021).
21. Rukhsar, & Upadhyay, S. K. Rice leaves disease detection and classification using transfer learning technique. *2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, 2151–2156 (2022). https://doi.org/10.1109/ICACITE53722.2022.9823596
22. Kathiresan, G., Anirudh, M., Nagharjun, M. & Karthik, R. Disease detection in rice leaves using transfer learning techniques. *J. Phys. Conf. Ser.* **1911**(1), 12004. https://doi.org/10.1088/1742-6596/1911/1/012004 (2021).
23. Mounika, P. & Govinda Rao, S. A study on deep learning and machine learning techniques on detection of Parkinson's disease. *E3S Web of Conferences*, 309, 1008 (2021). https://doi.org/10.1051/e3sconf/202130901008
24. Bhatheja, H. & Jayanthi, N. Detection of cotton plant disease for fast monitoring using enhanced deep learning technique. *2021 5th International Conference on Electrical, Electronics, Communication, Computer Technologies and Optimization Techniques (ICEECCOT)*, 820–825 (2021). https://doi.org/10.1109/ICEECCOT52851.2021.9708045

Author contributions

Conceptualization:I.S.Rahat, H. Ghosh,S.Dara and S.Kant ; Methodology:I.S.Rahat, H. Ghosh,S.Dara and S.Kant ; Validation:I.S.Rahat, H. Ghosh,S.Dara and S.Kant ; Writing-original Draft:I.S.Rahat, H. Ghosh,S.Dara and S.Kantt ; Writing-Review and Editing: All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.D. or S.K.

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